

Real Estate Price Prediction

<https://github.com/aiyashwary/real-estate-price-prediction>

OVERVIEW

A supervised regression case study that predicts house prices using the `kc_house_data` dataset, comparing performance with R^2 and MSE metrics.

IMPACT

Benchmarked six regression models with bagging/boosting and hyperparameter tuning to identify house prices.

TECHNICAL WALKTHROUGH

The project includes a Jupyter notebook, the `kc_house_data.csv` dataset, and a slide deck summarizing the project.

- i) Notebook: `ML_HOUSE_RENT_PREDICTION_CASE_STUDY IPYNB (1).ipynb`
- ii) Dataset: `kc_house_data.csv` (house attributes and prices)
- iii) Slides: `ML GROUP 8 (1).pptx`

Loaded the dataset, checked distributions and missing values, and prepared features for regression.

- i) Exploratory analysis with `pandas`, `matplotlib`, and `seaborn`.
- ii) Basic cleaning to ensure consistent numeric inputs.

Benchmarked multiple regression models to compare predictive performance.

- i) Linear Regression, KNN Regressor, Decision Tree Regressor

- ii) Random Forest, AdaBoost, Gradient Boosting Regressors

Applied bagging and boosting techniques, then used hyperparameter tuning to improve generalization.

- i) Model selection guided by R^2 and MSE metrics.

Compared model scores and selected the best-performing regressor based on error metrics.

Notebook can be run end-to-end with the dataset placed in the project root, producing charts and reports.

KEY DETAILS

1. Loaded and explored `kc_house_data.csv`, performing basic cleaning and feature analysis for data quality.
2. Trained Linear Regression, KNN Regressor, Decision Tree, Random Forest, AdaBoost, and Gradient Boosting Regressors.
3. Compared model performance using R^2 and MSE to select the best-performing regressor.
4. Applied bagging/boosting strategies and hyperparameter tuning to improve predictive accuracy.
5. Documented the workflow in a Jupyter Notebook and included presentation slides for project review.
6. Explicitly excludes logistic regression from the model comparison, noting it is a classification algorithm for price prediction.
7. Includes a PowerPoint slide deck alongside the notebook for presentation-ready project documentation.